E-LEARNING PLATFORM SALES AND PURCHASE ANALYSIS

1.Project Overview and Objective:

This project focuses on analysing an E-learning platform database using SQL. The data includes learners, courses, and purchase details.

The main objective of this project is to use SQL queries to extract meaningful insights such as learner spending, course demand, revenue trends, and category performance to support business decision-making.

2. Problem Statement

- To analyse learner purchasing behaviour
- To identify high-demand and low-demand courses
- To evaluate revenue generation across categories
- To understand country-wise performance
- To analyse monthly purchase trends

3.Table Description:

Learners Table

COLUMN NAME	DATATYPES	DESCRIPTION
learner_id	Integer	Unique ID of learner
full_name	String	Name of learner
country	String	Country of learner

Courses Table

COLUMN NAME	DATATYPES	DESCRIPTION
course_id	Integer	Unique ID of course
course_name	String	Name of course
category	String	Course category
unit_price	Decimal	Price of course

Purchases Table

COLUMN NAME	DATATYPES	DESCRIPTION
purchase_id	Integer	Unique ID of purchase
learner_id	Integer	Foreign key from learners
course_id	Integer	Foreign key from courses
quantity	Integer	Number of courses purchased
purchase_date	Date	Date of purchase

4.Schema Design:

1. Learners Table
   - Primary Key: learner_id
2. Courses Table
   - Primary Key: course_id
3. Purchases Table
   - Primary Key: purchase_id
   - Foreign Keys:
     - learner_id → references Learners(learner_id)
     - course_id → references Courses(course_id)

5. Analytical Queries

Q1. Display each learner's total spending (quantity × unit_price) along with their country.

```
select
   l.learner_id,
   l.full_name,
   l.country,
   ROUND(SUM(p.quantity * c.unit_price), 2) as total_spent from purchases p
join learners l on p.learner_id = l.learner_id
```

join courses c on p.course_id = c.course_id

group by l.learner_id, l.full_name, l.country order by total_spent desc;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

learner_id	full_name	country	total_spent
13	Olivia	UK	4400.42
41	Luca	Italy	4400.00
26	Zhang	China	4400.00
32	Omar	UAE	4000.00
17	Ethan	USA	3500.80
2	Rahul	India	3401.20
20	Harper	USA	3200.40
44	Yuki	Japan	3200.00
35	Hassan	UAE	3200.00
10	Ashok	USA	3100.40
8	Arun	India	3001.55
5	Velu	China	2801.20
23	Ravi	India	2800.42
29	Kim	Korea	2800.00

Insights:

A small group of learners contributes the highest total spending, indicating high-value customers. Most learners have moderate spending, showing varied purchasing behaviour across users.

Q2. Find the top 3 most purchased courses based on total quantity sold.

select

c.course_name,

SUM(p.quantity) as total_quantity

from purchases p

join courses c on p.course_id = c.course_id

group by c.course_name

order by total_quantity desc

limit 3;

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

course_name	total_quantity
SQL Basics	4
Python Basics	3
Advanced SQL	2

Insights:

The top courses have significantly higher quantity sold, indicating strong learner demand. These courses are key drivers of engagement and revenue on the platform.

Q3. Show each course category's total revenue and the number of unique learners who purchased from that category.

```
select
    c.category,
    ROUND(SUM(p.quantity * c.unit_price), 2) as total_revenue,
    COUNT(distinct p.learner_id) as unique_learners from purchases p
join courses c on p.course_id = c.course_id
group by c.category
order by total_revenue desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_revenue	unique_learners
Data	33805.37	15
Cloud	15701.30	6
AI	11301.72	4
Development	8402.65	5
Visualization	8001.52	4
Security	7900.00	3
Management	5900.00	3
System	4100.00	2
Design	4000.42	2
Tools	800.32	1
Soft Skills	800.00	1

Insights:

Certain categories generate higher total revenue and attract more learners. This shows that technical and career-oriented categories are more popular among users.

Q4. List all learners who have purchased courses from more than one category.

```
select
    l.learner_id,
    l.full_name,
    COUNT(distinct c.category) as category_count
from purchases p
join learners l on p.learner_id = l.learner_id
join courses c on p.course_id = c.course_id
group by l.learner_id, l.full_name
having COUNT(distinct c.category) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

learner_id	full_name	category_count
3	John	2

Insights:

Some learners purchase courses from multiple categories, indicating interest in diverse skill development. These learners are more engaged and valuable to the platform.

Q5. Identify courses that have not been purchased at all.

```
select
    c.course_id,
    c.course_name
from courses c
left join purchases p on c.course_id = p.course_id
where p.purchase_id is null;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_id	course_name
106	Excel Advanced
147	Leadership
148	Interview Prep
149	Resume Building
150	Career Guidance

Insights:

Some courses have no purchase records, indicating low demand or lack of awareness. These courses may require improvement in content or marketing strategies.

6.Key Insights

- High Demand Courses: Courses with the highest total quantity sold are the most popular among learners and contribute significantly to overall revenue.

- Low Demand Courses: Some courses have no purchase records, indicating low visibility or lack of interest from learners.
- Top Performing Categories: Certain categories generate higher revenue, showing strong demand in those areas.
- Learner Behaviour: A few learners have purchased multiple courses across different categories, contributing more to total sales.
- Country-wise Performance: Some countries show higher purchase activity and revenue compared to others.
- Purchase Trends: Sales vary over time, with certain periods showing higher purchase activity.

7.Recommendations

- Focus marketing efforts on high-demand courses and top-performing categories to maximize revenue.
- Improve visibility of low-demand courses through promotions, discounts, or updated content.
- Provide special offers or loyalty programs to high-value learners.
- Target marketing campaigns in countries with high purchase activity.
- Plan promotions during peak purchase periods to increase overall sales.

8. Conclusion

SQL analysis helped in understanding learner behaviour, course demand, and revenue patterns. The project demonstrates how SQL can be effectively used for data analysis and business insights generation.